Session 5

Data wrangling

Filtering

The filter verb extracts particular observations based on a condition. In this exercise you'll filter for observations from a particular year.

```
library(gapminder)
library(dplyr)

# Filter the gapminder dataset for the year 1957

gapminder %>%
  filter(year == 1957)
```

Arranging

`arrange()` orders the rows of a data frame by the values of selected columns.

```
library(dplyr)

arrange(.data, ...)
```

Exercise

In one sequence of pipes on the gapminder dataset:

- `filter()` for observations from the year 1967, Americas continent and population greater than 10000000
- `arrange()` in descending order of the life expectancy

The mutate verb

The `mutate` verb is used to create a new variable or change value variable from a data set.

```
library(dplyr)

mutate(.data, ...)
```

Combining filter, mutate, and arrange

In this exercise, you'll combine all three of the verbs you've learned to find the countries with the highest life expectancy, in months, in the year 2007.

In one sequence of pipes on the gapminder dataset:

`filter()` for observations from the year 2007,
`mutate()` to create a column `lifeExpMonths`, calculated as `12 * lifeExp`
`arrange()` in descending order of that new column

Visualizing with ggplot2

`ggplot2` is a system for declaratively creating graphics. You provide the data, tell `ggplot2` how to map variables to aesthetics, what graphical primitives to use, and it takes care of the details.

`ggplot()` initializes a ggplot object. It can be used to declare the input data frame for a graphic and to specify the set of plot aesthetics intended to be common throughout all subsequent

```
library(ggplot2)

ggplot(data, aes(x = _ , y = _ )) + geom_
```

Variable assignment

Comparing variables

Log scales

Break

15 minutes

Additional aesthetics

Putting the x-axis on a log scale

You previously created a scatter plot with population on the x-axis and life expectancy on the y-axis. Since population is spread over several orders of magnitude, with some countries having a much higher population than others, it's a good idea to put the x-axis on a log scale.

```
ggplot(gapminder_1952, aes(x = pop, y = lifeExp)) +
  geom_point() + scale_x_log10()
```

Putting the x- and y- axes on a log scale

Suppose you want to create a scatter plot with population on the x-axis and GDP per capita on the y-axis. Both population and GDP per-capita are better represented with log scales, since they vary over many orders of magnitude.

```
ggplot(gapminder_1952, aes(x = pop, y = gdpPercap)) +
geom_point() +
scale_x_log10() +
scale_y_log10()
```

Adding color to a scatter plot

In this lesson you'll learn how to use the color aesthetic, which can be use to show which continent each point in a scatter plot represents.

Adding size and color to a plot

In the last exercise, you'll create a scatter plot communicating information about each country's population, life expectancy, and continent. Now you'll use the size of the points to communicate even more.

Faceting

Creating a subgraph for each continent

You'll learn to use faceting to divide a graph into subplots based on one of its variables, such as the continent.

Faceting by year

Now that you're able to use faceting, you can create a graph showing all the country-level data from 1952 to 2007, to understand how global statistics have changed over time.

Exercise

In one sequence of pipes on the gapminder dataset:

- `filter()` for observations from the year 1972,
- `ggplot()` to create a geometry of points with axis x (`pop`) and axis y(`lifeExp`)
- add scale log in axis x
- add faceting for `continent` variable